

Session 14 Task

B1:

Answer: The console prints 3.99, but the database is NOT updated.

Explanation: FirstAsync loads the Student and EF tracks it. Changing s.Gpa modifies only the in-memory object. Without SaveChangesAsync(), no UPDATE is sent to SQL Server.

B2:

Answer: Each instructor is printed with Courses.Count = 0.

Explanation: Courses is initialized with new(), so it is an empty list, not null. Lazy loading is disabled and Include() was not used, so related courses are never loaded.

B3:

Answer: The database is not updated after SaveChangesAsync().

Explanation: AsNoTracking() disables change tracking. EF Core does not detect the GPA modification, so SaveChangesAsync() has nothing to save. This is dangerous because the code looks correct but silently does nothing.

QC:

C2:

```
F:\iti\NET Projects and Tasks\ x + v - □ x
warn: 8/3/2026 19:08:54.545 CoreEventId.SensitiveDataLoggingEnabledWarning[10400] (Microsoft.EntityFrameworkCore.Infrastructure)
Sensitive data logging is enabled. Log entries and exception messages may include sensitive application data; this mode should only be enabled during development.
info: 8/3/2026 19:08:54.987 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
Executed DbCommand (26ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT TOP(1) [s].[Id], [s].[FullName], [s].[Gpa], [s].[YearOfStudy]
FROM [Students] AS [s]
WHERE [s].[FullName] = N'Nada Samir'
Nada's Gpa: 3.90
Nada's Gpa: 4.00
|
```

Changed in C# object only

bo.Students [Data] x StudentPortalContext.cs Lab_14_Task.md [Preview]

Max Rows: 1000

	Id	FullName	YearOfStudy	Gpa
	1	Yara Adel	2	3.5
	2	Omar Hesham	3	2.8
	3	Nada Samir	1	3.9
	4	Kareem Fouad	4	3.2
	NULL	NULL	NULL	NULL

Didn't change in DataBase bec we didn't invoke context.savechanges()

C3:

```
using (var context = new StudentPortalContext())
{
    var nada = await context.Students.FirstOrDefaultAsync(s => s.FullName == "Nada Samir");
    Console.WriteLine($"Nada's Gpa: {nada.Gpa:F2}");
    nada.Gpa = 4.0;
    Console.WriteLine($"Nada's Gpa: {nada.Gpa:F2}");
    await context.SaveChangesAsync();
}
```

dbo.Students [Data] X Lab_14_Task.md [Preview] Program.cs

Max Rows: 1000

	Id	FullName	YearOfStudy	Gpa
▶	1	Yara Adel	2	3.5
	2	Omar Hesham	3	2.8
	3	Nada Samir	1	4
	4	Kareem Fouad	4	3.2
⚙	NULL	NULL	NULL	NULL

F:\iti\NET Projects and Tasks\ X + v

warn: 8/3/2026 19:20:13.218 CoreEventId.SensitiveDataLogging (SensitiveDataLoggingStructure)
Sensitive data logging is enabled. Log entries should only be enabled during development.

info: 8/3/2026 19:20:13.655 RelationalEventId.CommandExecuted (RelationalCommandExecuted)
Executed DbCommand (22ms) [Parameters=[], CommandType='Text', CommandText='SELECT TOP(1) [s].[Id], [s].[FullName], [s].[YearOfStudy], [s].[Gpa] FROM [Students] AS [s] WHERE [s].[FullName] = N'Nada Samir']

Nada's Gpa: 3.90
Nada's Gpa: 4.00

info: 8/3/2026 19:20:13.788 RelationalEventId.CommandExecuted (RelationalCommandExecuted)
Executed DbCommand (24ms) [Parameters=[@p1='Nada Samir', @p0='4.00'], CommandType='Text', CommandText='SET IMPLICIT_TRANSACTIONS OFF; SET NOCOUNT ON; UPDATE [Students] SET [Gpa] = @p0 OUTPUT 1 WHERE [Id] = @p1;']

Changes saved to DataBase

C4:

The screenshot shows a Visual Studio IDE with two main components: a SQL table viewer and a console window.

SQL Table Viewer:

- Tab: `dbo.Students [Data]`
- Max Rows: 1000
- Table Data:

	Id	FullName	YearOfStudy	Gpa
▶	1	Yara Adel	2	3.5
	2	Omar Hesham	3	2.8
	3	Nada Samir	1	4
	4	Kareem Fouad	4	3.2
	5	Youssef Ezzat	2	3.6

Console Window:

- Tab: `F:\iti\NET Projects and Tasks\`
- Log Output:

```
warn: 8/3/2026 19:26:36.494 CoreEventId.SensitiveDataLoggingEnabled(Structure)
    Sensitive data logging is enabled. Log entries and exception mode should only be enabled during development.
info: 8/3/2026 19:26:36.932 RelationalEventId.CommandExecuted[20]
    Executed DbCommand (48ms) [Parameters=[@p0='Youssef Ezzat' CommandType='Text', CommandTimeout='30']
    SET IMPLICIT_TRANSACTIONS OFF;
    SET NOCOUNT ON;
    INSERT INTO [Students] ([FullName], [Gpa], [YearOfStudy])
    OUTPUT INSERTED.[Id]
    VALUES (@p0, @p1, @p2);
id=5
|
```

C5:

The screenshot shows a database application window with a table named `dbo.Students [Data]`. The table has five columns: `Id`, `FullName`, `YearOfStudy`, and `Gpa`. The data is as follows:

Id	FullName	YearOfStudy	Gpa
1	Yara Adel	3	3.5
2	Omar Hesham	3	2.8
3	Nada Samir	1	4
4	Kareem Fouad	4	3.2
5	Youssef Ezzat	3	3.6

Below the table, a console window displays the following logs:

```
warn: 8/3/2026 19:30:08.813 CoreEventId.SensitiveDataStructure)
    Sensitive data logging is enabled. Log entries
    mode should only be enabled during development.
info: 8/3/2026 19:30:09.271 RelationalEventId.CommandExecuted DbCommand (29ms) [Parameters=[], CommandText="SELECT [s].[Id], [s].[FullName], [s].[Gpa], [s].[YearOfStudy] FROM [Students] AS [s]"]
info: 8/3/2026 19:30:09.391 RelationalEventId.CommandExecuted DbCommand (15ms) [Parameters=[@p1='5', @p0=3], CommandText="SET IMPLICIT_TRANSACTIONS OFF;
SET NOCOUNT ON;
UPDATE [Students] SET [YearOfStudy] = @p0
OUTPUT 1
WHERE [Id] = @p1;"]
```

C6: removed

The screenshot shows a Visual Studio IDE with two main windows. The top window, titled 'dbo.Students [Data]', displays a table with 4 rows and 5 columns: Id, FullName, YearOfStudy, and Gpa. The bottom window, titled 'F:\iti\NET Projects and Tasks\...', shows a console output with several log messages.

	Id	FullName	YearOfStudy	Gpa
▶	1	Yara Adel	3	3.5
	2	Omar Hesham	3	2.8
	3	Nada Samir	1	4
	4	Kareem Fouad	4	3.2

```
warn: 8/3/2026 19:36:53.017 CoreEventId.Sensitive
ructure)
    Sensitive data logging is enabled. Log entr
mode should only be enabled during development.
info: 8/3/2026 19:36:53.429 RelationalEventId.Com
Executed DbCommand (24ms) [Parameters=[], C
SELECT [s].[Id], [s].[FullName], [s].[Gpa],
FROM [Students] AS [s]
info: 8/3/2026 19:36:53.550 RelationalEventId.Com
Executed DbCommand (18ms) [Parameters=[@p0=
SET IMPLICIT_TRANSACTIONS OFF;
SET NOCOUNT ON;
DELETE FROM [Students]
OUTPUT 1
WHERE [Id] = @p0;
```

D3:

The screenshot shows a Visual Studio IDE with a C# migration file and the Package Manager Console output.

Code File: `StudentPortalContext.cs`

```

10  /// <inheritdoc />
11  protected override void Up(MigrationBuilder migrationBuilder)
12  {
13      migrationBuilder.AlterColumn<string>(
14          name: "FullName",
15          table: "Students",
16          type: "nvarchar(100)",
17          maxLength: 100,
18          nullable: false,
19          oldClrType: typeof(string),
20          oldType: "nvarchar(max)");
21
22      migrationBuilder.AlterColumn<string>(
23          name: "CourseName",
24          table: "Courses",
25          type: "nvarchar(150)",
26          maxLength: 150,
27          nullable: false,
28          oldClrType: typeof(string),
29          oldType: "nvarchar(max)");

```

Package Manager Console Output:

```

8/3/2026 20:43:43.730 CoreEventId.SensitiveDataLoggingEnabledWarning[10400] (Microsoft.EntityFrameworkCore.Infrastructure)
Sensitive data logging is enabled. Log entries and exception messages may include sensitive application data; this mode should only be
enabled during development.
An operation was scaffolded that may result in the loss of data. Please review the migration for accuracy.
To undo this action, use Remove-Migration.
PM>

```

1. Up performs an AlterColumn operation on FullName.
2. nullable: false oldNullable: true
3. any row that contain vlues agiants the rule

The screenshot shows SQL Server Enterprise Designer with a table design and the corresponding T-SQL script.

Table Design:

Name	Data Type	Allow Nulls	Default
Id	int	☐	
FullName	nvarchar(100)	☒	
YearOfStudy	int	☐	
Gpa	float	☐	

Keys (1): PK_Students (Primary Key, Clustered: Id)

Check Constraints (0)

Indexes (0)

Foreign Keys (0)

Triggers (0)

T-SQL Script:

```

1  CREATE TABLE [dbo].[Students] (
2      [Id] INT IDENTITY (1, 1) NOT NULL,
3      [FullName] NVARCHAR (100) NULL,
4      [YearOfStudy] INT NOT NULL,
5      [Gpa] FLOAT (53) NOT NULL,
6      CONSTRAINT [PK_Students] PRIMARY KEY CLUSTERED ([Id] ASC)

```

```
StudentPortalConsole
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const int ID = 34;
var threshold = 2.5 + (ID % 3) * 0.4;
var ex_val = ID % 5 + 3;
var gpa = 3.0 + ((ID % 7) * 0.1);

using (var context = new StudentPortalContext())
{
    var st = new Student
    {
        FullName = null!,
        YearOfStudy = 2,
        Gpa = gpa
    };

    context.Students.Add(st);
}

F:\iti\NET Projects and Tasks\ x + -
Executed DbCommand (47ms) [Parameters=[@p0=NULL (Nullable = false) (Size = 110), @p1='3.6', @p2='2'], CommandType=
'Text', CommandTimeout='30']
SET IMPLICIT_TRANSACTIONS OFF;
SET NOCOUNT ON;
INSERT INTO [Students] ([FullName], [Gpa], [YearOfStudy])
OUTPUT INSERTED.[Id]
VALUES (@p0, @p1, @p2);
fail: 8/3/2026 21:12:01.342 CoreEventId.SaveChangesFailed[10000] (Microsoft.EntityFrameworkCore.Update)
An exception occurred in the database while saving changes for context type 'StudentPortalConsole.StudentPortalCon
text'.
Microsoft.EntityFrameworkCore.DbUpdateException: An error occurred while saving the entity changes. See the inner
exception for details.
----> Microsoft.Data.SqlClient.SqlException (0x80131904): Cannot insert the value NULL into column 'FullName', tab
le 'ITI_StudentPortal.dbo.Students'; column does not allow nulls. INSERT fails.
at Microsoft.Data.SqlClient.SqlConnection.OnError(SqlException exception, Boolean breakConnection, Action`1 wra
```


F:

The image shows a Visual Studio Code editor window with a C# file named `StudentPortalConsole.Migrations.AddInstructorCourseRelationship.cs`. The code defines a partial class `AddInstructorCourseRelationship` that inherits from `Migration`. It implements the `Up` method, which drops the `AssignedCourseName` column from the `Instructors` table and adds an `InstructorId` column to the `Courses` table. The code includes comments explaining the data loss risk of the `DropColumn` operation and a note about migrating existing data first.

```
4
5 namespace StudentPortalConsole.Migrations
6 {
7     /// <inheritdoc />
8     public partial class AddInstructorCourseRelationship : Migration
9     {
10         /// <inheritdoc />
11         protected override void Up(MigrationBuilder migrationBuilder)
12         {
13             migrationBuilder.DropColumn(
14                 name: "AssignedCourseName",
15                 table: "Instructors");
16             // The DropColumn operation destroys data.
17             // It permanently removes the AssignedCourseName column,
18             // so every value stored in that column is lost.
19
20             //solution a real project would migrate the existing data first
21             migrationBuilder.AddColumn<int>(
22                 name: "InstructorId",
23                 table: "Courses",
24                 type: "int",
25                 nullable: false,
26                 defaultValue: 0);
27 }
```

The Package Manager Console at the bottom displays a warning: "An operation was scaffolded that may result in the loss of data. Please review the migration for accuracy. To undo this action, use Remove-Migration."

